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(21218) Roll No. .....

BCA-V Sem.

18024

B. C. A. Examination, Dec. 2018

Numerical methods

(BCA-504)

(New Course)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions. Calculator is allowed.

Section-A

(Very Short Answer Questions)

Attempt all the *five* questions. Each question carries 3 marks. 3×5=15

1. Find $\sqrt{12}$ by applying Newton-Raphson's method.

2. Prepare a divided difference table for the following data :

x	1	2	4	7	12
$f(x)$	22	30	82	106	216

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3. Find the first derivative of $f(x)$ at $x = 0.4$ from the following table :

x	0.1	0.2	0.3	0.4
$y = f(x)$	1.1051	1.2214	1.3498	1.4918

4. Solve :

$$5x - y - 2z = 142$$
$$x - 3y - z = -30$$
$$2x - y - 3z = 5$$

by Gauss's elimination method.

5. Given that :

$$\frac{dy}{dx} = \frac{y-x}{y+x}, \quad y(0) = 1,$$

find $y(0.1)$ by Picard's method.

Section-B

(Short Answer Questions)

Attempt any *two* questions out of the following three questions. Each question carries 7½ marks. 7½×2=15

6. Given $f(0)=16.35$, $f(5)=14.88$, $f(10)=13.59$, $f(15)=12.46$ and $f(x) = 14.00$, find x.

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7. Estimate the sale for 1966 using the following data :

Year	Sales (in thousand)
1931	12
1941	15
1951	20
1961	27
1971	39
1981	52

8. Find the root of $x^2 - 5x + 2 = 0$ correct to five decimal places by Newton-Raphson's method.

Section-C

(Detailed Answer Questions)

Attempt any *three* questions out of the following five questions. Each question carries 15 marks. 15×3=45

9. Using Runge-Kutta method, find an approximate value of y for $x = 0.2$ if $\frac{dy}{dx} = x + y^2$, given that $y = 1$ when $x=0$ and $h = 0.1$.

10. Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's '3/8' rule. Hence obtain the approximate value of π.

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11. Solve by Gauss-Seidel method of iteration, the equations :

$$10x + y + z = 12$$
$$2x + 10y + z = 13$$
$$2x + 2y + 10z = 14.$$

12. Using Stirling formula, find $f(28)$ from the following table :

$f(20)=49225$, $f(25)=48316$, $f(30)=47236$, $f(35)=45926$, $f(40)=44306$.

13. Find the real root of the equation $x \log_{10} x - 1.2 = 0$. Correct to five places of decimal.

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(21218) Roll No. .....

BCA-V Sem.

18024

B. C. A. Examination, Dec. 2018

Numerical methods

(BCA-504)

(New Course)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions. Calculator is allowed.

Section-A

(Very Short Answer Questions)

Attempt all the *five* questions. Each question carries 3 marks. 3×5=15

1. Find $\sqrt{12}$ by applying Newton-Raphson's method.

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x	1	2	4	7	12
$f(x)$	22	30	82	106	216

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$$x - 3y - z = -30$$
$$2x - y - 3z = 5$$

by Gauss's elimination method.

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$$\frac{dy}{dx} = \frac{y-x}{y+x}, \quad y(0) = 1,$$

find $y(0.1)$ by Picard's method.

Section-B

(Short Answer Questions)

Attempt any *two* questions out of the following three questions. Each question carries 7½ marks. 7½×2=15

6. Given $f(0)=16.35$, $f(5)=14.88$, $f(10)=13.59$, $f(15)=12.46$ and $f(x) = 14.00$, find x.

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Section-C

(Detailed Answer Questions)

Attempt any *three* questions out of the following five questions. Each question carries 15 marks. 15×3=45

9. Using Runge-Kutta method, find an approximate value of y for $x = 0.2$ if $\frac{dy}{dx} = x + y^2$, given that $y = 1$ when $x=0$ and $h = 0.1$.

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$$2x + 2y + 10z = 14.$$

12. Using Stirling formula, find $f(28)$ from the following table :

$f(20)=49225$, $f(25)=48316$, $f(30)=47236$, $f(35)=45926$, $f(40)=44306$.

13. Find the real root of the equation $x \log_{10} x - 1.2 = 0$. Correct to five places of decimal.

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(21119)
B.C.A.-V Sem.

18024

B.C.A. Examination, November-2019
NUMERICAL METHODS
(BCA-504)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from *all* sections as per instructions. Calculator is allowed.

Section–A
(Very Short Answer Questions)

Note : Attempt all *five* questions. Each question carries *3 marks*. Very short answer is required not exceeding 75 words. 5×3=15

1. Evaluate the following for h = 1 :
$E = e^{hD}$

2. Construct a divided difference table for the following data :

x	3	5	9	15
y	2	14	38	74

3. Write the formula for Simpson's 3/8th rule.

4. Write the formula for Runge-Kutta method for 4th order.

5. Perform two iterations of Picard's method to find an approximate solution of the initial value problem :

$$\frac{dy}{dx} = x - y, \quad y(0) = 1$$

Section–B
(Short Answer Questions)

Note : Answer any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required not exceeding 200 words. 2×7½=15

6. Using Euler's method find an approximate value of y for x = 1, considering h = 0.5, given that

$$\frac{dy}{dx} = x + y, \quad \text{and } y(0) = 1$$

7. Apply Newton's divided difference method to obtain an interpolatory polynomial for the following data:

x	3	5	7	9	11	13
f(x)	31	51	17	19	90	110

8. Find the first two derivatives of f(x) at x = 1 from the following table :

x	−2	−1	0	1	2	3	4
f(x)	104	17	0	−1	8	69	272

Section-C

(Detailed Answer Questions)

Note : Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 3×15=45

9. Derive Newton– Raphson's method to find a root of the equation f(x) = 0. Prove that this method has Quadratic Convergence.

10. Solve the following system of linear equations using Gauss-Seidel method :

$$10x + 3y + 7z = 41$$
$$3x + 20y + 17z = 101$$
$$x + 19y + 23z = 201$$

perform three iterations.

11. Define Lagrange's interpolation formula. Obtain Lagrange's interpolatory for the following data :

x	1	3	5	7	10
f(x)	13	31	25	37	101

12. State Runge-Kutta method of second order. Using Runge-Kutta method of fourth order find the values of y (0.2), y (0.4) and y (0.6) for the following initial value

$$\frac{dy}{dx} = x^3 - y^3$$

Write Condition that y (0) = 1.

13. Evaluate

$\int_0^1 \frac{dx}{1+x^2}$ by using

(a) Trapezoidal rule

(b) Simpson's 1/3 rule

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(20221) Roll No.

BCA.-V Sem.

18024

B.C.A. Examination, Dec. 2020

Numerical Methods

(BCA-504)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions. Calculator is allowed.

Section-A

(Very Short Answer Questions)

Note : Answer all the **five** questions. Each question carries **3** marks. $3\times5=15$

1. Find the real root of the equation:

   $f(x)= x^3-x-1=0$

2. Define operators E, Δ and ∇. Also obtain relation between E and Δ.

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3. Given $\frac{dy}{dx} = \frac{y-x}{y+x}$ with $y=1$ for $x=0$. Find $y(0.3)$ by Euler's method taking $h=0.1$.

4. Define Simpson's three-eight rule for Numerical integration.

5. What do you understand by Gauss's eliminations method?

Section-B

(Short Answer Questions)

Note : Attempt any **two** questions out of the following **three** questions. Each question carries **7½** marks.

$7\frac{1}{2}\times2=15$

6. Use Picard's method to approximate y when $x=0.2$, given that $y=1$ when $x=0$ and $\frac{dy}{dx} = x-y$.

7. Find the value of y when $x=10$ for the following table:

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x	5	6	9	11
y	12	13	14	16

8. Using Bessel's formula, find y (25) from the following data:

x	20	24	28	32
y	24	32	35	40

Section-C

(Detailed Answer Questions)

Note : Attempt any **three** questions out of the following **five** questions. Each question carries **15** marks. 15×3=45

9. Apply Gauss-Seidal iteration method to solve the equations.

$20x+y-2z=17$

$3x+20y-z=-18$

$2x-3y+20z=25$

10. Use Runge-Kutta method of fourth order, to find y (0.2) for the equation.

$\frac{dy}{dx} = \frac{y-x}{y+x}$ y(0)=1, take h=0.2

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11. Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using.

(a) Trapezoidal rule

(b) Simpson's ⅓ rule.

12. By using Newton-Raphson's method, find the root of $x^4-x-10=0$ which is nearer to x=2 correct to three places of decimal. Also obtain the rate of Convergance of Newton-Raphson's method.

13. (a) The following value of the function f(x) for values of x are given:

f(1)=4, f(2)=5, f(7)=5, f(8)=4.

Find the value of f(6) and also the value of x for which f(x) is maximum or minimum.

(b) Apply Lagrange's formula to find the cubic polynomial which includes the following values of x and y_x:

x	0	1	4	6
y_x	1	−1	1	−1

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(20222) Roll No. .....

BCA-V Sem.

18024 (CV-III)

B.C.A. Examination, Dec.-2021

Numerical Methods

(BCA-504)

Time : 1½ Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** Sections as per instructions. Calculator is allowed.

Section-A

(Very Short Answer Questions)

Note : Attempt any **two** questions. Each question carries **7.5** marks.

2×7.5=15

1. Find a root of the eq $f(x)= x^3-4x-9=0$ using the bisection method in four iterations.

2. Find the form of the function from following given data:

x :	0	1	2	3	4
f(x):	3	6	11	18	27

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3. Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by Trapezoidal Rule.

4. Use Euler's Method with h=0.1 to find the solution of $\frac{dy}{dx} = x^2 + y^2$, $y(0)=0$ in the range $0 \le x \le 0.5$

5. Solve by Gauss-elimination method.

$2x+y+4z=12$

$8x-3y+2z=23$

$4x+11y-z=33$

Section-B

(Short Answer Questions)

Note : Attempt any **one** question out of the following three questions. Each question carries **15** marks. 1×15=15

6. By means of Newton's divided difference formula find the value of f(8) and f(15) from the following table.

x:	4	5	7	10	11	13
f(x):	48	100	294	900	1210	2028

7. From the given table. Find $\frac{dy}{dx}$ at x=1.2.

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x	y
1.0	2.7183
1.2	3.3201
1.4	4.0552
1.6	4.9530
1.8	6.0496
2.0	7.3891

8. Using Picard's method of successive approximation obtain a solution upto fourth approximation of the equation.
$\frac{dy}{dx} = y + x, y(0) = 1$

Section-C

(Detailed Answer Questions)

Note : Attempt any **two** questions out of the following five questions. Each question carries **22.5** marks.

https://www.ccsustudy.com $2 \times 22.5 = 45$

9. Find a real root of the equation $x^3 - x^2 - 2 = 0$ by False Position Method.

10. Interpolate by mean of Gauss's formula the population for the year 1936, given the following table.

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Year (x)	Population (y) (in thousand)
1901	12
1911	15
1921	20
1931	27
1941	39
1951	52

11. Find the value of the integral $\int_0^1 \frac{dx}{1+x}$ by using Simpson's $\frac{1}{3}$ and $\frac{3}{8}$ rule.

12. Using Runge-Kutta method of fourth order solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at $x = 0.2, 0.4$.

13. Find the solution of the system by Gauss-Seidel Method.

$83x + 11y - 4z = 95$

$7x + 52y + 13z = 104$

$3x + 8y + 29z = 71$

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(21222)

BCA-V Sem.

(Printed Pages 4)

Roll No. .....

18024

B.C.A. Examination, Dec. 2022

Numerical Methods

(BCA-504)

Time : Three Hours] *[Maximum Marks :75*

Note : Attempt questions from **all** sections as per instructions. Calculator is allowed.

Section - A

(Very Short Answer Questions)

Note : Attempt **all** questions. Each question carries **3** marks. $3\times5=15$

1. Obtain a real root of the equation $f(x)=x^3-x-1=0$ using bisection method correct to three decimal places..

2. Define forward difference. Make forward difference table for the following ordered pairs. : (x_0, y_0), (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4), (x_5, y_5), (x_6, y_6).

P.T.O.

3. Find the value of $\frac{dy}{dx}$ at x=2.2 for the following ordered pairs :

(1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891), (2.2, 9.0250).

4. Solve the following system of linear equations using Gaurs elimination method. :

$2x+y+z=10$

$3x+2y+3z=18$

$x+4y+9z=16$

5. Solve the equation $\frac{dy}{dx}=x+y^2$, y=1 when x=0 using picard's method.

Section - B

(Short Answer Questions)

Note : Attempt any **two** questions out of the following three questions. Each question carries **7.5** marks. $7.5\times2=15$

6. Find a real root of the equation :

$f(x) = x^3-2x-5=0$ using method of false position.

7. Find the missing term in the following table.

x	Y
0	1
1	3
2	9
3	–
4	81

8. Evaluate $I=\int_0^{\pi/2} \sqrt{\sin x}\ dx$ using Simpson's $\frac{1}{3}$ rule with $h = \frac{\pi}{12}$

Section - C

(Detailed Answer Questions)

Note : Attempt any **three** questions out of the following **five** question Each question carries **15** marks.

$3\times15=45$

9. Use Newton-Raphson method to find a real root of the equation $4e^{-x} \sin x-1=0$ correct to 3 decimal places, given that the root lies between 0 and 0.5

10. Using Lagrange's inter polation formula, find the form of the function y(x) from the following table.

x	0	1	3	4
y	–12	0	12	24

11. Use simpson's $\frac{3}{8}$ –rule to evaluate $\int_0^1 \frac{1}{1+x} dx$ with $n=\frac{1}{6}$

12. Solve the system of linear equations given below by Gauss's siedel Iterative method

$10x+2y+z=9$

$2x+20y-2z=-44$

$-2x+3y+10z=22$

13. Using fourth - order Ranga - Kutta method find y(0.2) and y(0.4) correct to four decimal places, given $\frac{dy}{dx}$ 1+y2 where y(0)=0

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(21223) Roll No. .....

B.C.A.-V Sem.

18024

B.C.A. Examination, Dec.-2023

Numerical Methods

(BCA-504)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** sections as per instructions. Calculator is allowed.

Section-A

Note : Attempt all the **five** questions. Each question carries **3** marks. 3×5=15

1. Using Newton-Raphson method, find an iterative formula to compute $\sqrt[3]{N}$, where N is a positive number.

2. Prove that $E\nabla = \nabla E = \Delta$, where E is shift operator , ∇ is backward difference operator & Δ is forward difference operator.

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3. Find the first derivative of the function given below at the point x = 1.2

x	1	2	3	4	5
f(x)	0	1	5	6	8

4. Solve the following equations by using Gauss-elimination method :

$$x - y + z = 1$$
$$-3x + 2y - 3z = -6$$
$$2x - 5y + 4z = 5$$

5. Perform two iterations of Picard's method to find an approximate solution of the initial value problem

$y' = x+y^2$; $y(0) = 1$.

Section-B

Note : Attempt any **two** questions out of the following three questions. Each question carries 7.5 marks. 2×7.5=15

6. Using method of False position, find a positive root of the equation

$x^3 - 4x+1 = 0$.

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7. Using Newton's forward interpolation formula, find the cubic polynomial which takes on the following values

x	0	1	2	3	4
y	-1	0	13	50	123

8. Using Simpson's $\left(\frac{1}{3}\right)^{rd}$ rule, find the value of $\int_{-1}^{5} f(x)\,dx$ given that :

x	1	2	3	4	5
f(x)	10	50	70	80	100

Section-C

Note : Attempt any **three** questions out of the following five questions. Each question carries 15 marks. 3×15=45

9. Find a positive root of the equation $2x=3+\cos x$ by Bisection method.

10. Using Lagrange's interpolation formula, find the value of y corresponding to x=2 from the following table :

x	0	1	3	4
y	5	6	50	105

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11. Evaluate $\int_{0}^{1} \frac{dx}{1+x^2}$ by using Simpson's $\left(\frac{3}{8}\right)^{th}$ rule.

12. Solve by Gauss-Seidel method

$$3x+y+z=1$$

$$x+3y-z=11$$

$$x-2y+4z=21.$$

13. Apply Runge-Kutta method fourth order to find an approximate value of y when x=0.2, given that

$\frac{dy}{dx}=x+y^2$ & y= 1 when x = 0

(take h= 0.1) 6.5181

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